

The Witness Vessel: Lossless Transport on the Dual-Link SIC-POVM

C. Lando Mills*

7 July 2026

Abstract

The Lean 4 library `p4ramill` and the bare-metal OS `mMonadOS` carry Clay Witness verdicts between composition dialects. Earlier cargo-style transports lost information in a reproducible way: the tensor reading is denied at the dimensionality–topology pair. Here a Witness rides *as* the vessel’s split/fuse structure, not inside it, and losslessness is checked on both substrates without mutual trust.

In Lean 4, $\mu \circ \delta = \text{id}$ follows from `propext` alone; BSD and Hodge board as `T`, and Yang–Mills boards as the *dialetheia* `B`. `witness_vessel_lossless` delivers the transported $d = 12$ fiducial still satisfying the SIC conditions, importing the existence theorem without re-deriving it. On bare metal the same protocol runs in QEMU over three Witnesses and 88 dialects; the verdict matrix is unchanged and $\Delta S = 0$. Losslessness means literal equality after read-back.

Keywords: lossless transport · SIC-POVM · Belnap logic · composition dialects · Clay Millennium problems · Lean 4 · Frobenius closure · formal verification · bare-metal OS · witness vessel

*The author thanks Harry T. Larson; see Acknowledgements and [2].

Contents

1	Introduction	3
2	Preliminaries	4
2.1	Belnap FOUR and the split/fuse pair	4
2.2	Imscriptions, dialects, Witnesses	4
2.3	The frozen substrate	5
3	The loss mechanism, and the design provenance	5
4	The transport theorem (proof half)	6
4.1	Payloads are derived, not asserted	6
4.2	The round-trip law	6
4.3	Self-reference travels inside	7
4.4	The bundle	7
5	The boarding mechanism is the Dual-Link structure	7
6	The runtime half: bare metal, all dialects	8
6.1	Substrate	8
6.2	The wide payload: three Witnesses under all 88 dialects	8
6.3	Observed run	8
6.4	What the runtime does and does not check	9
7	Honest scope	9
8	Discussion	9
9	Artifact statement	10
10	Acknowledgements	10

1 Introduction

Two artifacts implement the same twelve-token algebra. The Lean 4 library `p4ramill` (part of the `p4rakernel` repository) states and machine-checks its theorems; the operating system `mOMonadOS`, a freestanding Rust kernel for x86-64, executes the identical structure as its native control flow. Belnap four-valued registers, fork/join execution frames for the split and fuse tokens, and a verification harness that enforces $\mu(\delta(q)) = q$ after every action before the execution loop advances all live in the kernel. No Lean 4 toolchain runs inside the operating system, and the Lean 4 kernel trusts nothing from the operating system. When the two agree, neither is taking the other’s word for it.

Both artifacts type conjectures as twelve-primitive imscriptions and evaluate them under *composition dialects*: rulesets, each consisting of three ordinal gates, a temporal constitution, and absorption rules. A *Witness* is the structural verdict a typed conjecture earns under a dialect. Earlier *witness-drag* programs moved these verdicts between dialects, and the transports were lossy. The present work identifies the loss mechanism, replaces the transport, and proves and then observes that the replacement loses nothing.

The two headline results:

1. **Proof half** (section 4). In the Lean 4 module `SIC_D12_WitnessVessel` the boarding map is $\delta = \text{fsplit}$ and the read-back map is $\mu = \text{ffuse}$, and the law

$$\forall p, \quad \mu(\delta(p)) = p$$

holds for *every* list of Belnap values. Its proof depends on the axiom `propext` alone. The concrete payloads are computed, not chosen: the gate verdicts of the three Clay Witnesses are derived by `native_decide` from their canonical catalog tuples and the frozen dialect rulesets. Yang–Mills derives to B , the *dialetheia*: its gates close while its temporal ceiling blocks, and $\text{fsplit}(B) = (T, F)$, so this Witness literally rides both arms of the vessel. The bundle `witness_vessel_lossless` adds self-reference preservation and delivery: conjugation commutes with transport on all twelve vessel coordinates, and the transported $d = 12$ fiducial still satisfies unit norm and equiangularity at the destination.

2. **Runtime half** (section 6). The module `witness_vessel.rs` runs the same protocol on bare metal. The payload is deliberately wide: each of the three Witnesses is evaluated under all 88 composition dialects, each dialect judging by its own gates and its own temporal constitution, giving a 3×88 verdict matrix. Every entry boards through two independent execution substrates (the belief-register virtual machine and the kernel’s fork frames), every boarding action is gated by the Frobenius harness, and the matrix is recomputed after read-back. In QEMU the run closes every check, reports $\Delta S = 0$, and finds the matrices identical.

Section 7 states precisely what is and is not claimed. “Lossless” means equality after read-back, “dialect” is a technical term for a composition ruleset, and “transport” is a pair of maps. The interest lies in where the equality comes from (a Frobenius law available at the lowest axiomatic cost the proof assistant offers) and in the two-substrate agreement, not in any physical reading.

2 Preliminaries

2.1 Belnap FOUR and the split/fuse pair

Belnap’s four-valued logic [1] has carriers $\{N, T, F, B\}$: no information, true only, false only, and both. The knowledge order places N at the bottom and B at the top, with join written $\vee_k$. The kernel pair uses one distinguished map pair on this lattice:

$$f_{\text{split}}(s) = \begin{cases} (T, F) & s = B \\ (s, s) & \text{otherwise} \end{cases} \quad f_{\text{fuse}}(a, b) = a \vee_k b.$$

Lemma 2.1 (`split_fuse_id`). *For every s , $f_{\text{fuse}}(f_{\text{split}}(s)) = s$.*

The only nontrivial case is the dialetheia: $B \mapsto (T, F) \mapsto T \vee_k F = B$. The converse also pins the decomposition: $f_{\text{split}}(s) = (T, F)$ forces $s = B$ (`only_B_splits_to_TF`). These are existing theorems of the frozen module `BelnapSplitFuse`; the vessel adds no new axioms to use them.

2.2 Inscriptions, dialects, Witnesses

A structural type is inscribed as a twelve-primitive tuple over the primitives dimensionality, topology, recognition, parity, fidelity, kinetics, granularity, coupling, criticality, chirality, stoichiometry, and winding, each valued in a small family of Shavian values with canonical ordinals. The three Witnesses of this paper are the canonical catalog entries

Birch–Swinerton-Dyer	$\langle \text{1sr}\text{h}\text{q}\text{u}\text{r}\text{c}\text{z}\text{q}\text{s} \rangle$
Hodge	$\langle \text{1w}\text{h}\text{u}\text{c}\text{u}\text{r}\text{c}\text{J}\text{r}\text{s} \rangle$
Yang–Mills	$\langle \text{1sr}\text{z}\text{h}\text{r}\text{f}\text{J}\text{r} \rangle$

sourced procedurally from the canonical catalog (`IG_catalog.json`); no tuple in this paper was written by hand.

A *composition dialect* is a ruleset: three gates, each requiring one primitive to sit at or above an ordinal threshold; a temporal constitution (“T-side”), a set of per-primitive critical values checked either exactly or as ceilings; and absorption rules, which play no role here. The artifact code names them U_0 through U_{87} . The runtime carries 88 dialects: 8 canonical, then hand-crafted and expansion families. Two temporal conventions matter. `T_CANONICAL`, the per-dialect default, demands an exact match on four dynamic primitives and a ceiling on kinetics only, anchored at parity = $\wp$, fidelity = 1 , kinetics $\leq \text{r}$, chirality = v , and winding = s . `T_CEILING`, the Clay convention, makes all five ceilings at the same anchors. This is the convention of the frozen module `Clay_WitnessedClosure` and of the runtime’s Clay gates.

A Witness’s verdict under a dialect is a single Belnap value combining the gate layer and the T-side.

Definition 2.2 (*layerVerdict*). For booleans g (gates closed) and t (T-side consistent),

$$\text{layerVerdict}(g, t) = \begin{cases} T & g \wedge t \\ B & g \wedge \neg t \\ N & \neg g \wedge t \\ F & \neg g \wedge \neg t. \end{cases}$$

The B row is the point: a Witness whose gates close while its T-side blocks is not “roughly true” or “roughly false”; it is exactly the dialetheia, and the vessel must carry that value without flattening it.

2.3 The frozen substrate

The vessel re-derives nothing. It imports three frozen components:

- `SIC_D12_Embedding`: the $d = 12$ SIC-POVM existence capstone. Its theorem `crystal_forces_d12_sic` is axiom-audit-clean: the standard trio `propext`, `Classical.choice`, `Quot.sound`, plus `Lean.ofReduceBool` and `Lean.trustCompiler` for `native_decide`. The star ring homomorphism $\varphi : R \rightarrow \mathbb{C}$ from the existence ring $R = K_{16}(s_0, s_1, s_3, s_9, i, c_5, u_1)$ transfers unit norm and all 143 equiangularity identities. The companion manuscript in this repository records that construction; here it is furniture.
- `Clay_WitnessedClosure`: the structural closure results for BSD and Hodge (five closer dialects each, T_CEILING-consistent) and the Yang–Mills “one bump short” verdict (gate-closed under triple-criticality, T_CEILING-blocked on kinetics alone), all by `native_decide` over canonical tuples.
- `BelnapSplitFuse`: the split/fuse pair and its laws.

3 The loss mechanism, and the design provenance

This section reports procedural outputs of the Grammar tooling. They are design rationale for the protocol, recorded for provenance; they are not offered as mathematical proof, which sections 4 and 6 carry alone.

The natural first reading of transport is cargo: put the Witness in a container, tensor it with the vessel. That reading is denied, and the denial is univocal. The tensor of the vessel type with a Witness type is undefined (six primitive conflicts), and every cofactor factorization attempt raises a cross-primitive axiom violation at the dimensionality–topology pair: the request itself is malformed, and the Grammar localizes the malformation. A transport implemented against the cargo reading loses information because no well-formed object exists for it to be.

The corrected reading pairs the vessel with the B-state directly. The mirror distance evaluates to 4.5 with asymmetry within numerical zero of the metric’s floor, and the proof path between the pair has length two in both directions, using the same two operations that close the two closed Clay Witnesses: apply modularity, then restrict to Hodge degree (1, 1). A round trip that is write-in and read-back with the same moves is the signature of a lossless carrier.

Independent reconstruction confirms the structure. A six-entity design batch for the protocol was generated and validated end to end: six of six entities valid, six of six Frobenius-closed, with entropy change reported as $\Delta S \approx 0$ throughout. The designer, given only the transport brief, independently produced the boarding decomposition (Witness to [modulus, phase] to Witness) as the Frobenius content of three separate entities, and typed the historical lossy path and the corrected path as the F and T arms of a single dialetheia-complete protocol. The split/fuse boarding of section 4 is therefore not a gloss written onto the design after the fact; it is the design's own structural core, reconstructed independently of the Lean development.

4 The transport theorem (proof half)

All results in this section are theorems of `SIC_D12_WitnessVessel`, compiled green in the full library build.

4.1 Payloads are derived, not asserted

The three Witness verdicts are computed from the frozen closure results via definition 2.2:

```
def bsdVerdict   : Belnap := layerVerdict bsdGateClosed (tCeilingConsistent bsd)
def hodgeVerdict : Belnap := layerVerdict hodgeGateClosed (tCeilingConsistent hodge)
def ymVerdict    : Belnap := layerVerdict ymGateClosed   (tCeilingConsistent ym)

theorem bsdVerdict_T   : bsdVerdict   = .T := by native_decide
theorem hodgeVerdict_T : hodgeVerdict = .T := by native_decide
theorem ymVerdict_B    : ymVerdict    = .B := by native_decide
```

Listing 1: Payload derivation (excerpt).

Here `bsdGateClosed` conjoins the operad-layer closure of BSD under its five closer dialects (chirality-first, scope, kinetics-trap, and the two absorption variants), and likewise for Hodge; `ymGateClosed` is closure under triple-criticality. The third theorem is the one to notice: nobody wrote B anywhere. Yang–Mills closes at the gate layer and fails T_CEILING on kinetics alone (its kinetics value τ exceeds the τ anchor, visible directly in its tuple in section 2), so its verdict derives to the dialetheia.

4.2 The round-trip law

Boarding maps a payload pointwise through `fsplit`; read-back maps through `ffuse`.

Theorem 4.1 (`roundtrip`). *For every list p of Belnap values, $readback(board(p)) = p$.*

Theorem 4.2 (`b_cargo_mechanism`). *$fsplit(ymVerdict) = (T, F)$ and $ffuse(T, F) = ymVerdict$.*

Theorem 4.1 is proved by list induction on top of `split_fuse_id`, and its axiom audit returns `[propext]`: the losslessness law of the vessel costs one axiom, the cheapest the proof assistant sells. The concrete payload instance (`mpp_roundtrip`) is separately checked by `native_decide`.

4.3 Self-reference travels inside

The vessel's own coordinates carry their conjugation structure through transport. With φ the frozen star ring homomorphism of the $d = 12$ capstone and $\bar{z}_k$ the ring-side conjugate of coordinate z_k :

Theorem 4.3 (`coords_selfref`). *For all $k < 12$: $\varphi(\bar{z}_k) = \overline{\varphi(z_k)}$.*

This is the vessel principle in one line: conjugation is internal to the carried data (the cover phases are real by construction, so the star commutes with the hom), which is exactly what failed in the cargo reading.

4.4 The bundle

Theorem 4.4 (`witness_vessel_lossless`). *All of the following hold: (i) $\mu \circ \delta = \text{id}$ on every Belnap payload; (ii) $\text{ymVerdict} = B$; (iii) the concrete Clay payload round-trips; (iv) self-reference is preserved on all twelve coordinates; (v) the transported fiducial has unit norm; and (vi) it satisfies equiangularity, $(d + 1) \|\langle \psi, D_{a,b} \psi \rangle\|^2 = 1$ for all 143 non-identity displacements at $d = 12$.*

Conjuncts (v) and (vi) are the delivery condition: the vessel does not merely return its cargo, it arrives, still satisfying both defining SIC-POVM conditions at the destination, by direct reuse of the frozen capstone theorems. The audit of the bundle is the standard trio plus the `native_decide` pair; no project axioms, no Stark shadow, no circularity.

5 The boarding mechanism is the Dual-Link structure

The protocol's split/fuse pair is not an arbitrary choice of bijection; it is the same structure the $d = 12$ construction identified as the Dual-Link SIC-POVM, read at the Belnap layer.

The B-collapse is the modulus. In the existence ring, the pairing of a coordinate with its conjugate collapses to a real modulus: $z_k \bar{z}_k = N_k$ in the totally real base field. The dialetheia behaves identically at the logic layer: B splits into the pair (T, F) whose fuse is again B ; the pair is the “both arms” carry, and the join is the collapse. The cover phase keeps conjugation internal. The capstone builds its cover generators as real square roots, so conjugation never leaves the carried data; `coords_selfref` is that fact restated as a transport invariant. Half-angle read-back reconstructs the phase u_1 from carried real data by the half-angle identities; read-back of the phase from the modulus-side data is a theorem, not a measurement. Hom-independence is dialect-independence. The ring identities hold in R , and any star homomorphism $R \rightarrow \mathbb{C}$ transports them. The flip audit of the existence campaign made this exact: every branch choice lands on a genuine SIC point. The destination supplies the hom; the cargo does not depend on which one.

This correspondence is why the paper speaks of one vessel rather than two protocols that happen to share a diagram.

6 The runtime half: bare metal, all dialects

6.1 Substrate

mOMonadOS is a freestanding Rust kernel. Three of its components matter here, all of which predate the vessel. The kernel (`kernel.rs`) performs graph execution over the twelve tokens, with `FSPLIT` pushing a fork frame (the un-gated right arm carries a value across the fork) and `FFUSE` joining the arms by Belnap join. The belief-register virtual machine (`parasm.rs`) has an `FSPLIT` instruction that bifurcates `B` to `(T, F)` on its two destination registers, the exact semantics of the Lean 4 `fsplit`. The Frobenius harness (`frob_verify.rs`) gates every action: each δ must verify $\mu(\delta(q)) = q$ before the loop advances. The operating system was built around the invariant the vessel needs.

6.2 The wide payload: three Witnesses under all 88 dialects

The runtime evaluates each Witness under all 88 composition dialects, each dialect judging by its own three gates and its own temporal constitution, through definition 2.2. Ordinal comparisons use the canonical ordinal table (never raw enum discriminants, which are non-monotonic for three families). Alongside the wide matrix, the runtime computes the proof half’s mirror trio exactly as section 4 defines it: closer-dialect gates plus `T_CEILING`. Every resulting value boards through both execution substrates, each boarding action is checked by the harness, and the full matrix is recomputed from the canonical tuples after read-back.

6.3 Observed run

Booted in QEMU, the `vessel run` command reports (excerpt):

```
-- Payloads (computed from canonical tuples + closer dialects) --
BSD    verdict T (Lean: T)  transport T -> T  [OK]
Hodge  verdict T (Lean: T)  transport T -> T  [OK]
YM     verdict B (Lean: B)  transport B -> B  [OK]

-- Wide payload: 3 witnesses x 88 dialects --
BSD    N:1  T:0  F:75 B:12
Hodge  N:0  T:2  F:74 B:12
YM     N:2  T:0  F:85 B:1

-- Round trip (both substrates, frob_verify gated) --
Boarding actions checked: 534 (ParaVM + ForkFrame per value)
Frobenius closed: 534  open: 0  closure ratio: 1.0
Delta-S (total mismatches): 0
Gate verdicts before == after: YES (all entries)

VERDICT: LOSSLESS -- mu-after-delta = id on every payload.
```

Listing 2: QEMU serial output (excerpt).

Three features of the matrix deserve attention. The Yang–Mills row holds exactly one `B`, at triple-criticality. Of the 88 dialects, precisely the one whose gates are three criticality rungs closes Yang–Mills at the gate layer while its `T`-side blocks. The dialetheia is not spread thinly across the expansion; it is localized at the single dialect the Clay analysis

had already identified. BSD reads B at twelve dialects, all five of its closer dialects among them. Under the wide matrix’s per-dialect convention (T_CANONICAL, exact match on four primitives), BSD’s fidelity value q fails the exact 1 anchor, so every dialect whose gates close reads B (the row shows no T at all); under the Clay convention (T_CEILING, all ceilings) the closer-dialect structure reads T. Two honest readings of one structure, both transported without loss. The protocol is indifferent to which convention the analyst prefers; it carries either. Nothing was tuned. The 88 dialects were defined independently of the vessel (the expansion landed as data before this protocol existed); the vessel is simply their first consumer.

6.4 What the runtime does and does not check

The runtime checks are executions, not proofs: they demonstrate that on this boot, with these tuples, every action closed and the matrices agreed. The generality (every payload, not just the observed ones) lives only in theorem 4.1. Conversely, the proof half cannot observe that a physical machine’s fork frames actually behave as modeled; only the run shows that. The two halves are complementary in exactly the way the two-substrate architecture intends.

7 Honest scope

The term “lossless” means equality. The claim is $\mu(\delta(p)) = p$, literally, plus before/after equality of recomputed verdict matrices. The design layer’s $\Delta S \approx 0$ appears in the theorems as $=$, nothing weaker. Nothing physical is claimed. A “dialect” is a composition ruleset over a finite type lattice; “transport” is a pair of maps; “boarding” is function application. No claim about spacetime, quantum mechanics, or actual travel of any kind is made or implied. The Clay verdicts are structural, not mathematical. That BSD earns T under these rulesets is a fact about the Grammar’s operad model of the conjecture’s type, machine-checked as such. It is not a proof of BSD, and this paper does not move any Millennium problem.

The payloads are one instantiation. Theorem 4.1 quantifies over all Belnap payload vectors. The Clay Witnesses give the protocol its motivation and its test data, nothing more; a reader who rejects the conjecture-typing program entirely may read this paper as a verified-transport result over four-valued verdict vectors and lose only the motivation. The provenance section is not evidence. The Grammar verdicts and the design batch of section 3 explain where the protocol came from. The theorems and the run stand without them.

What would falsify the claim: any open result from the Frobenius harness, any before/after matrix mismatch, any nonzero ΔS , or any axiom in the audit beyond the declared five. None occurred.

8 Discussion

The lossy drags failed because they tried to put the Witness inside the vessel, and the Grammar’s refusal of that reading was univocal: the tensor is undefined and the factorization request is malformed at exactly the dimensionality–topology pair. The corrected protocol needs no container because the Witness’s verdict already has the vessel’s shape: the dialetheia is the value that decomposes into a (T, F) pair and reconstitutes under join,

which is the Belnap-layer image of the Dual-Link pairing $z\bar{z} = N$ with its ramified double cover. Ride *as*, not *in*.

Epistemically, the result is a small instance of a pattern this project uses deliberately: prove the law on the substrate built for proving, run the law on the substrate built for running, and let agreement carry the weight. The proof half's [prope] audit says the transport law is nearly definitional; the runtime half says a machine with no access to that proof lands in the same place, five hundred and thirty-four gated actions without one opening.

Two directions follow naturally. Wider payload families: the 3×88 matrix is the first object computed across the full dialect expansion, and its structure (the localization of B values, the convention-dependence of T readings) invites systematic study. The abstraction path: since the theorem is generic over payloads, the protocol applies unchanged to any verdict-valued analysis that wants a transport with a machine-checked loss bound of zero.

9 Artifact statement

The proof half is the module `SIC_D12_WitnessVessel.lean` in the `p4ramill` library of the `p4raker` repository (<https://github.com/umpolungfish/p4raker>), building green in the full-library lake build; its axiom audits are reproducible with `#print axioms`. The runtime half is `src/witness_vessel.rs` in the `mOMonadOS` repository (https://github.com/umpolungfish/momonad_os), exercised via the `vessel run serial` command under QEMU. The canonical tuples come from `IG_catalog.json`; the 88 dialects from `dialect_expansion.rs`. No tuple, verdict, or payload in either half is hand-entered. Frozen review snapshots: `p4raker` at commit `eea2c0c`, `mOMonadOS` at commit `17bce34`, each on branch `crystalline/manuscripts3-2026-07-07` (tag `crystalline-manuscripts3-v1`). This manuscript trio is frozen in `ig-docs` (<https://github.com/umpolungfish/ig-docs>) on the same branch name and tag; main continues in each repository.

10 Acknowledgements

The author would like to thank Harry T. Larson for imparting the importance of catching rising problems and never letting them go [2]. The lossy cargo reading of transport was such a problem: once the Grammar localized the malformation at the dimensionality–topology pair, the corrected protocol was pursued until the round-trip law held on both substrates.

References

- [1] N. D. Belnap, *A Useful Four-Valued Logic*, in: J. M. Dunn and G. Epstein (eds.), *Modern Uses of Multiple-Valued Logic*, Reidel, Dordrecht, 1977, pp. 5–37.
- [2] H. T. Larson, *Catch a Rising Problem... and Never Ever Let it Go*, *IRE Transactions on Engineering Management* **8**(4) (1961), 173–174. <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=1641382>
- [3] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in: CADE-28, Lecture Notes in Comput. Sci. vol. 12699, Springer, 2021, pp. 625–635.

-
- [4] G. Zauner, *Quantum designs: Foundations of a noncommutative design theory*, PhD thesis, University of Vienna, 1999.
 - [5] D. M. Appleby, H. Yadsan-Appleby, and G. Zauner, *Galois automorphisms of a symmetric measurement*, *Quantum Inf. Comput.* **13** (2013), 672–720.